SECTION 9: NUMERICAL ABILITY
Time -- 25 minutes
25 Questions (321 - 345)

NAME: _____
SCHOOL: _____
SIMULATED EXAM - SET B

For each question, decide which is the best answer choice, and darken the corresponding oval on the answer sheet.

321. Mario wants to buy a gadget whose marked price is P4,800 exclusive of tax. If there were 25% discount and 3% sales tax, how much would Mario pay for the gadget?
A. P3,615
B. P3,708
C. P3,724
D. P3,812

322. Mary was m years old when Barry was born. If Barry was b years old now, how old will Mary be in x years?
A. (b + m + x) years old
B. (m + x) years old
C. (b + x) years old
D. (bm + x) years old

323. The organizers of a certain concert were able to sold 300 tickets that cost P250 each. If they were to pay a 12% tax for the ticket sales, how much would their tax be?
A. P9,000
B. P9,250
C. P10,500
D. P11,200

324. A man from city A travels towards city B at an average speed of 36 kph for 1 hour and 35 minutes. What is the distance between the two cities?
A. 57 km
B. 48 km
C. 65 km
D. 42 km

325. If the salary of an employee increases by 10% every year, how many percent would the employee's salary increase in two years?
A. 19%
B. 20%
C. 21%
D. 22.5%

326. Charles works thrice as fast as John. Working alone, John alone can finish the job in 12 days. If Charles and John would work together, how long would it take them to finish the job?
A. 3 days
B. 6 days
C. 9 days
D. 12 days

327. James invested a certain amount of money at 12% per annum simple interest for t years. At the end of t years, James got back three times his original investment. What is the value of t?
A. 12 years
B. 16 years and 8 months
C. 20 years and 6 months
D. 25 years

328. In a 200-m race, Andrew beats Benzie by 8 seconds. On the contrary, if Andrew would give Benzie a head start of 32 meters, then they would finish the race at the same time. How long does Andrew take to run the 200-meter race?
A. 30 seconds
B. 42 seconds
C. 50 seconds
D. 58 seconds

329. If a 1200-km road is 3 cm long on the map, how long is a 800-km road on the map?
A. 1 cm
B. 1.5 cm
C. 2 cm
D. 2/3 cm

330. Out of ten students who took the Special Math quiz, three got a perfect score of 100. If the average of the scores of the ten students was 86, what was the average score of those who did not get a perfect score?
A. 72
B. 78
C. 80
D. 82

331. Volume varies inversely with pressure. The pressure inside a 500-ml air balloon is 3 atm. If the volume of the air balloon were increased by 1000 ml, what would happen to the pressure inside the air balloon?
A. would decrease by 1 atm
B. would increase by 1 atm
C. would decrease by 2 atm
D. would increase by 2 atm

332. Gregg and James are armchair assemblers. Gregg can assemble 24 armchairs in 1 hour. Working together, Gregg and James can assemble 24 armchairs in 20 minutes. How many armchairs can James make in 40 minutes?
A. 24
B. 32
C. 40
D. 48

333. Find the length of the base of a triangle whose area is 64 cm^2 and whose altitude is one-half of the base.
A. 16
B. $8\sqrt{2}$
C. 24
D. 32

334. A businessman invests part of $24,000 at 11% and the remainder at 8%. How much does the businessman invest at 8% if the two investments earn a total interest of $2,250?
A. $9,000
B. $11,000
C. $12,000
D. $13,000

335. A sales representative receives a basic salary of P12,000 per month and an additional 5% commission on all his sales. If his total sales for this month were P25,000, how much would his total earnings be?
A. P12,750
B. P13,125
C. P13,200
D. P13,250

336. A car travels 80 km/h and passes the welcome signage of a city at exactly 8:30 AM. Following the same route, a bus traveling 60 km/h passes the same welcome signage at 9:00 AM of the same day. At what time would the two vehicles be 90 km apart?
A. 10:30 AM of the same day
B. 10:40 AM of the same day
C. 11:00 AM of the same day
D. 11:30 AM of the same day

337. A canoe went 30 km upstream in three hours and took an hour shorter for the returned trip. Find the speed of the canoe in still water.
A. 2.5 kph
B. 7.5 kph
C. 9.5 kph
D. 12.5 kph

338. Nine workers can paint 3 houses in 2 days. How long would it take 12 workers to paint 4 houses?
A. 2 days
B. 3 days
C. 4 days
D. 5 days

339. If the price of rice increases by 20% and Aling Petra intends to spend only 5% more on rice, by how much percent should she reduce the quantity of rice that she buys?
A. 11%
B. 12.5%
C. 15%
D. 25%

340. If the mean of numbers 34, x, 96, 352 and 1045 is 323, then what is the mean of 54, 126, 392, 1165 and x?
A. 359
B. 365
C. 419
D. 424

341. The average of five consecutive integers starting with s is a. Find the average of nine consecutive integers starting with $s + 2$.
A. $a + 3$
B. $a + 4$
C. $a + 5$
D. $a + 6$

342. Tickets in a certain movie house cost P200.00 each. Senior citizens receive 20% discount. One day, the movie house sold 100 tickets and collected a total of P19,200.00. How many senior citizens went to the movie house that day?
A. 10
B. 15
C. 20
D. 30

343. A small inlet pipe can fill the tank in 6 hours while a larger inlet pipe can fill the same tank in 4 hours. On the other hand, a drain pipe can empty a full tank in 8 hours. How long would it take to fill an empty tank if all three pipes were open?
A. 1 11/13 hours
B. 2 1/5 hours
C. 2 4/7 hours
D. 3 3/7 hours

344. Anna has 20 coins consisting of one-peso and five-peso coins. What is the maximum value of her coins if the number of one-peso coins is at least twice the number of five-peso coins?
A. P36
B. P40
C. P44
D. P48

345. The sum of the third and fifteenth term of an arithmetic progression is 18. What is the sum of the first 17 terms of the arithmetic progression?
A. 108
B. 125
C. 141
D. 153